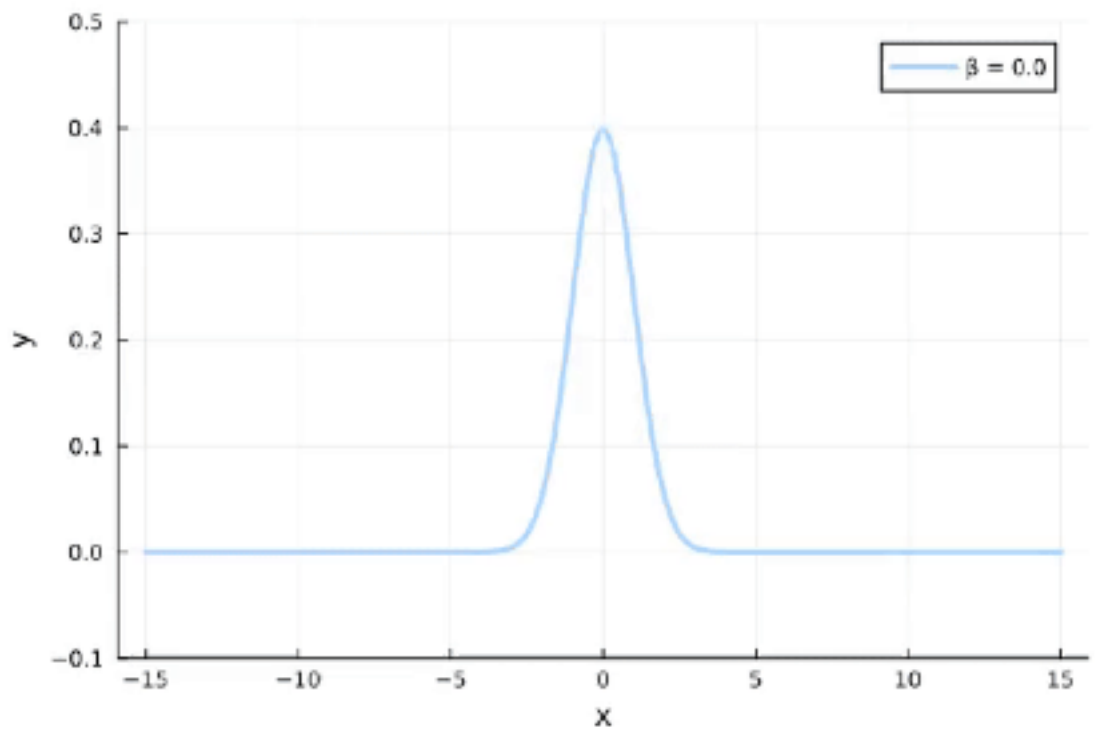
ANNEALING

22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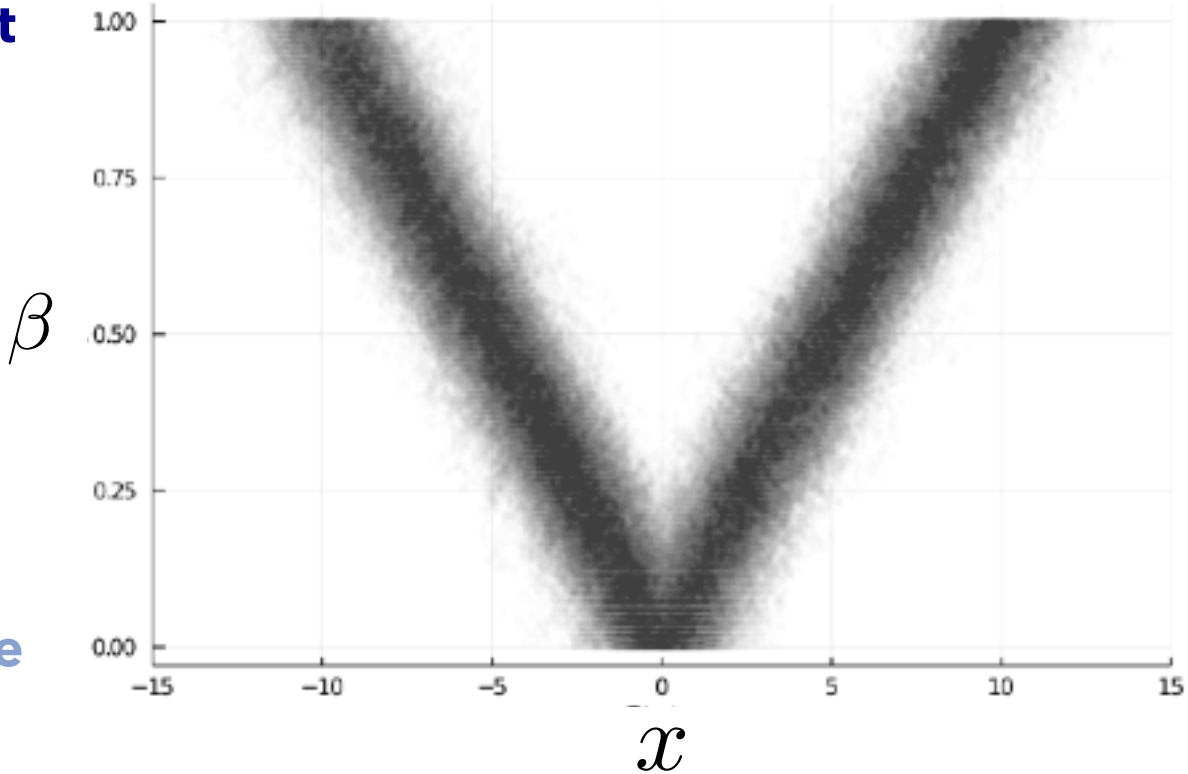
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
y
−15
−10
−5
0
5
10
15
x

- For $\beta \in [0,1]$ suppose we hav distributions π_β such that $\pi_0 = \eta$ and $\pi_1 = \pi$

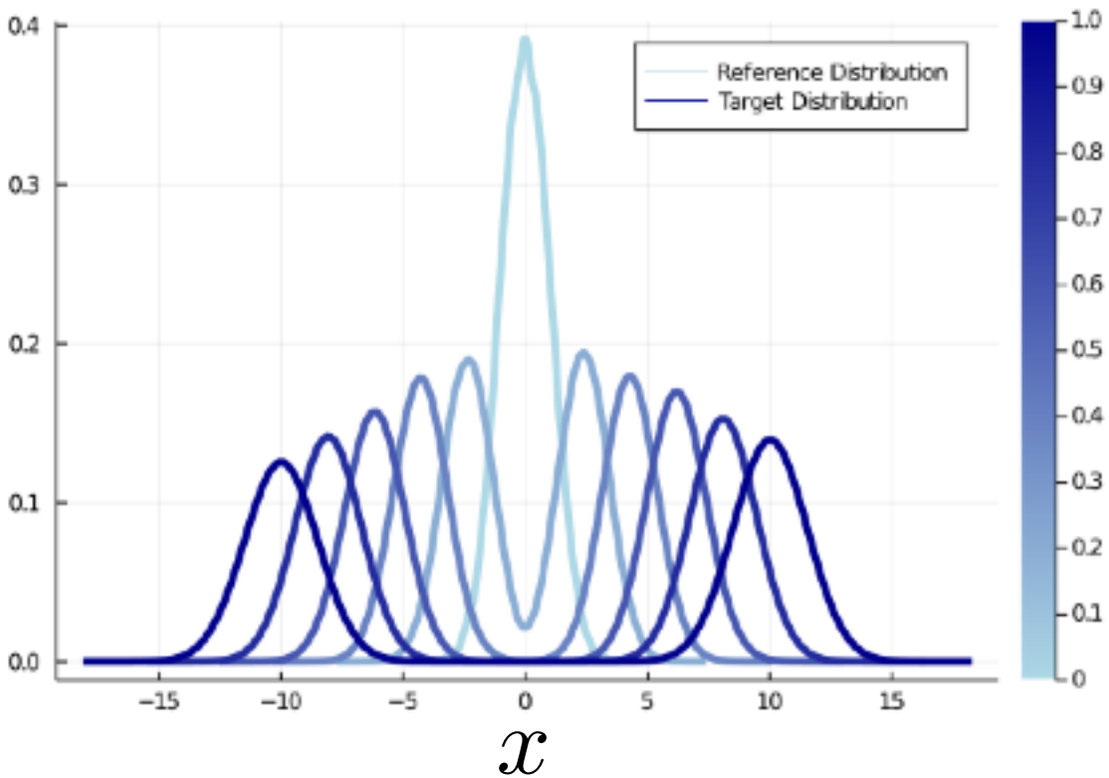
- π_β is the **annealing distribution** corresponding to the **annealing parameter** β

Target



Reference

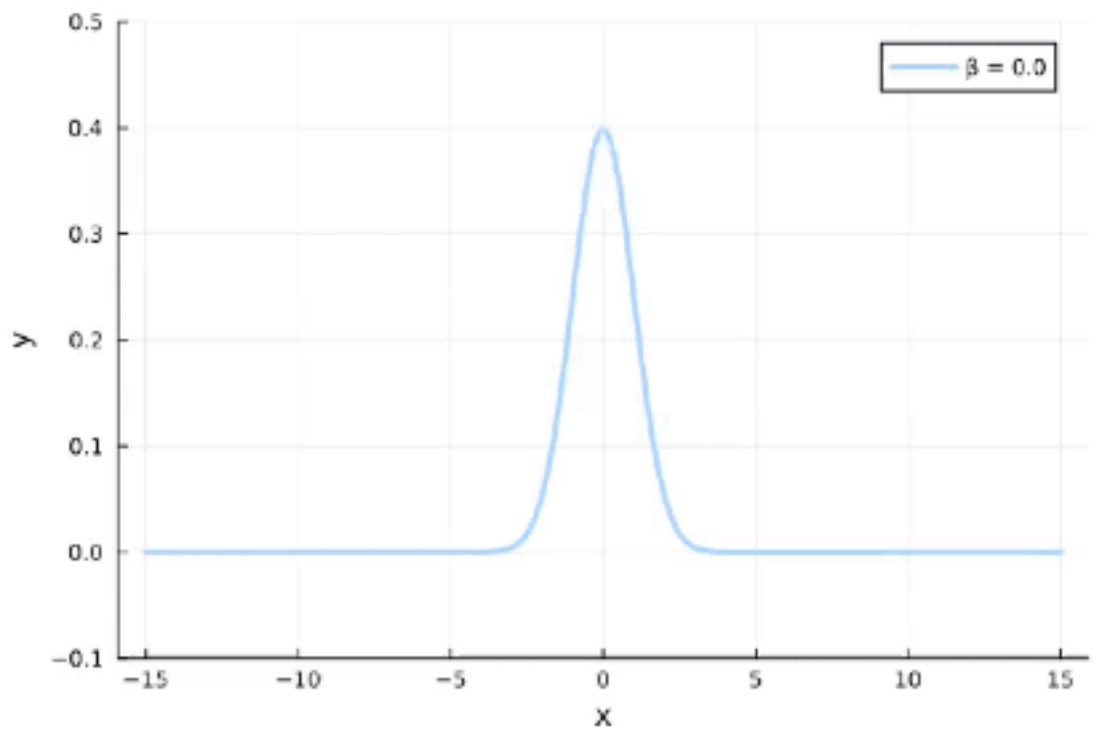
- We can construct our sequence of annealing distributions by setting $\pi_n = \pi_{\beta_n}$



- Note that π_{n-1} and π_n are close the schedule if fine and $\beta, \beta' \mapsto D(\pi_{\beta'} \| \pi_\beta)$ is continuous

- Where $\mathscr{B} = \beta_{0:N}$ is the **annealing schedule** satisfying:

$$0 = \beta_0 < \beta_1 < \cdots < \beta_N = 1$$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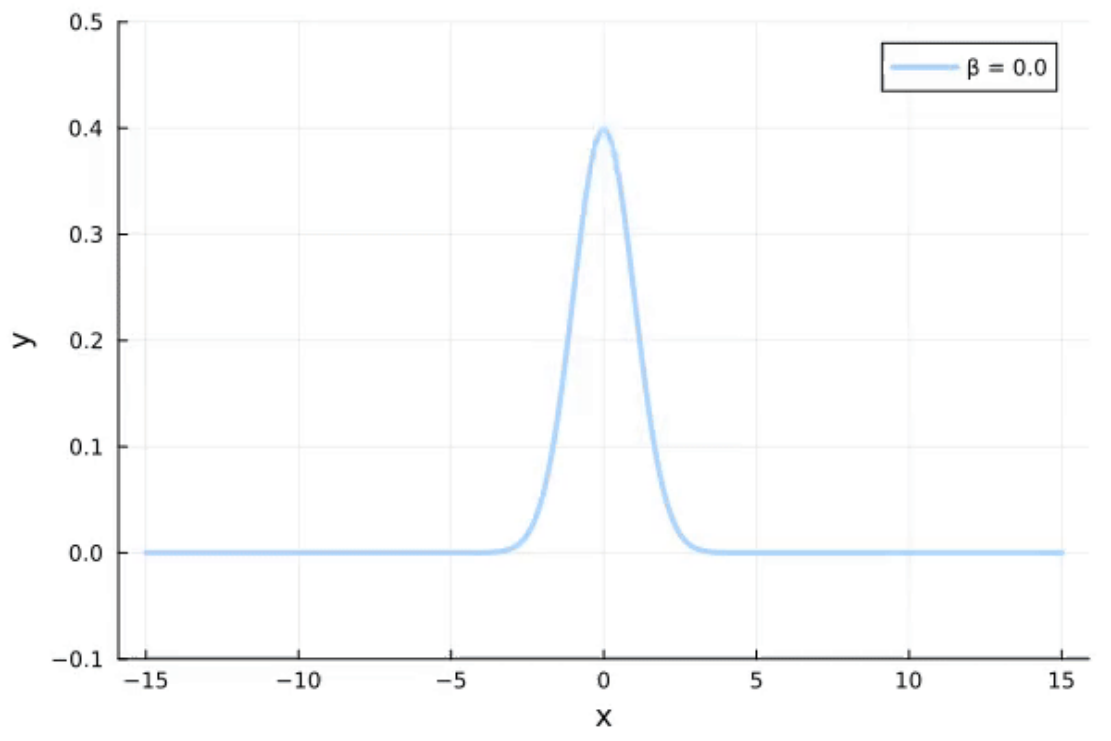
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ANNEALING

- For $\beta \in [0,1]$ suppose we hav distributions π_β such that $\pi_0 = \eta$ and $\pi_1 = \pi$
  - π_β is the **annealing distribution** corresponding to the **annealing parameter** β
- We can construct our sequence of annealing distributions by setting $\pi_n = \pi_{\beta_n}$
  - Where $\mathscr{B} = \beta_{0:N}$ is the **annealing schedule** satisfying:

$$0 = \beta_0 < \beta_1 < \cdots < \beta_N = 1$$

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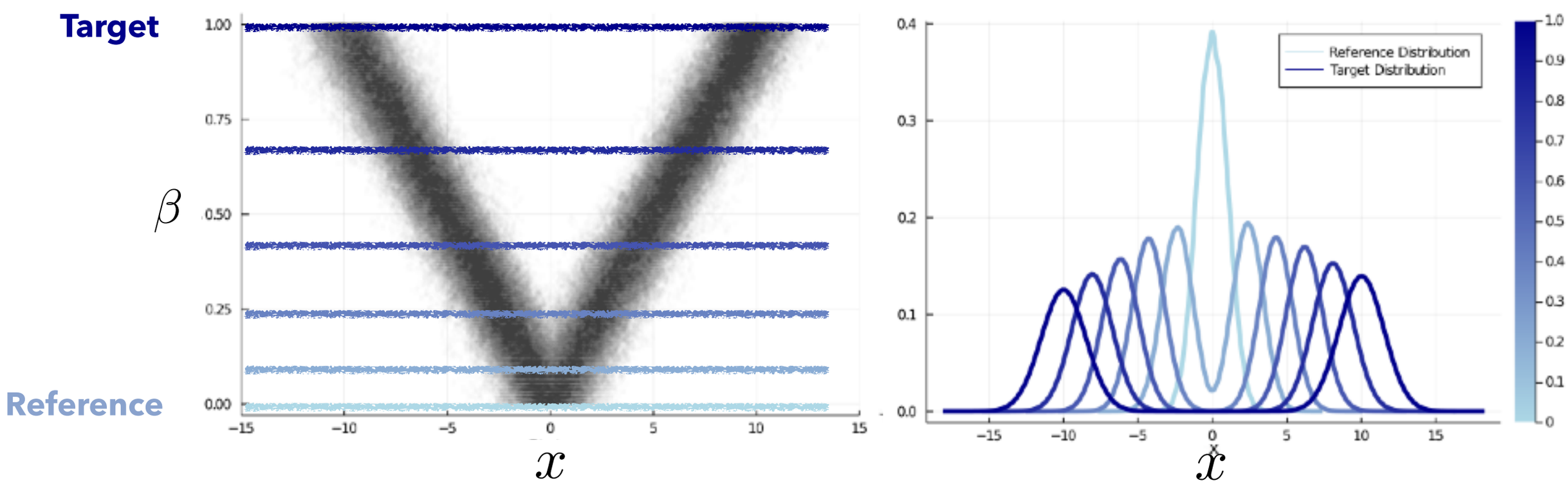

ANNEALING ALGORITHMS

- Annealing is **not** an algorithm but a platform to build algorithms on top of